

BSc in IT: Specialising in Data Science

IT3021: Data Warehousing
and Business Intelligence

Lecture 06

QA in Data Warehousing

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Importance of Testing

- Data drives critical business decisions; thus, testing the data warehouse data integration process is essential
- Data comes from numerous sources; hence, data sources affect data quality.
- Source data history, business rules, or audit information may change; testing must be an ongoing process
- In the ETL process, data flows through a pipeline before reaching the data warehouse. Thus, entire ETL pipeline must be tested to ensure each type of data is transformed or copied as expected
- Data must be available on a timely manner and be consistent
- Only relevant people/applications must have access to data in the data warehouse and also only the minimum required level of access should be given
- BI layer must ensure the correctness of data presented; this is the front end of data warehousing layer for business users

Types of Testing in Data Warehousing

- ETL/ELT testing
- DW performance testing
- DW security testing
- BI testing

ETL/ELT Testing

- Ensures ETLs run as intended
- ETL in different stages must be tested
- Data completeness/quality testing
 - Data at every entry/exit point must be tested
 - If not, the data that is harmed, altered inadequately or lost on the way, will results in misleading insights and cause faulty business decisions

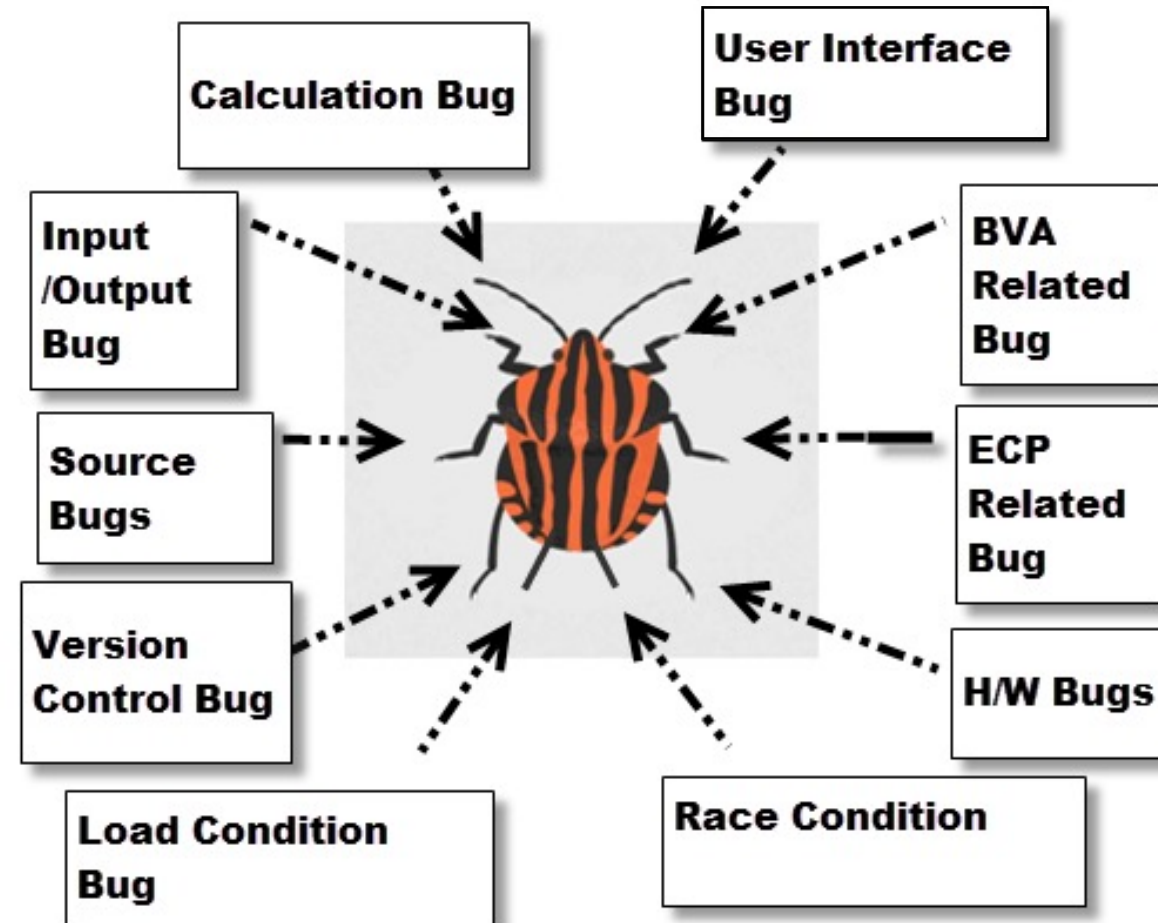
ETL/ELT Testing Types

- Metadata testing: ensures that the table definitions conform to the data model and application design specifications
 - Tests: table structure check, data type check, data length check, and index/constraint check, etc.
- Data completeness testing: validates that all the expected source data has been successfully loaded to the target
 - Tests: compare and validate counts, aggregates (min, max, sum, average), and actual data between the source and destination, rejected record report
- Data quality (correctness) testing: ensures data quality issues are handled as intended
 - Tests: NULL replacements checks, duplicate check, etc.
- Data transformation testing:
 - White box testing: review the transformation logic from the mapping design document and the ETL code. This is used to create test cases. (data enrichment, business logic, record create/modify dates, SCD records, etc.)
 - Black box testing: examines the functionality of an application without looking at internal structures for transformation testing; this involves reviewing output of ETL process against defined test cases

ETL/ELT Testing Types

- ETL regression testing: validates that the ETL produces the same output for a specific input before and after the change
- Incremental ETL testing: verifies that updates on the sources are getting loaded into the target system correctly
- ETL integration testing: end-to-end testing of data flows through the ETL process from all of the source systems to target application (dw/semantic layer)
- ETL performance testing: end-to-end testing to ensure that the all steps in the ETL process are working with expected data volumes
 - One pitfall of this testing method is the lack of actual data to emulate appropriate volumes

Types of ETL Bugs



Types of ETL Bugs

Types of Bug	Description
User interface bugs/cosmetic bugs	Related to GUI of application Font style, font size, colours, alignment, spelling mistakes, navigation and so on
Boundary Value Analysis (BVA) related bug	Minimum and maximum values
Equivalence Class Partitioning (ECP) related bug	Valid and invalid type
Input/Output bugs	Valid values not accepted Invalid values accepted
Calculation bugs	Mathematical errors Final output is wrong

Types of ETL Bugs

Types of Bug	Description
Load Condition bugs	Does not allows multiple users Does not allows customer expected load
Race Condition bugs	System crash & hang System cannot run client platforms
Version control bugs	No logo matching No version information available This occurs usually in regression testing
H/W bugs	Device is not responding to the application
Help Source bugs	Mistakes in help documents

Automation of ETL Testing

- General methodology of ETL testing:
 - SQL scripting
 - Do “eyeballing” of data.
- These approaches to ETL testing are time-consuming, error-prone and hardly provide complete test coverage
- To accelerate, improve coverage, reduce costs, improve defect detection ratio of ETL testing in production and development environments, automation is the needed
- Further, Some testing needs to be done iteratively with each ETL execution.

Challenges in ETL Testing

- Different types of formatted data
- Data of Heterogeneous sources
- Huge volume of data
- Lack of testing tools
- Lack of testing knowledge/ and experience
- Lack of requirements clarity
- Missing values in large volume of data
- Outdated data warehousing tools
- DW systems could have different architectures
- Complex data warehousing architectures with many ETL tools and components
- Automation of testing will impact on ETL execution performance

Data Warehouse Performance Testing

- Detects possible bottlenecks
- Ensures data warehouse handles the increasing number of user requests
 - Performing various operations simultaneously
- Ensures data warehouse handles the growing data volume from data sources

Data Warehouse Security Testing

- Ensures the correctness of role-based access
- Ensures the correct work of data encryption and decryption implementations
- Ensures the validity of data backup operations (if applicable)

Testing in Business Intelligence Layer

- Ensures OLAP operations work correctly
 - Roll-up
 - Drill-down
 - Slicing
 - Dicing
 - Pivot
- Ensures data presented in reports and dashboards are accurate
- Ensures rendering time of reports and dashboards are in agreement with requirements
- Ensures the permissions set in reports and dashboards are in line with the requirements

Testing Process

- Business requirements gathering
 - Helps to test the accuracy of the data model, define business flow requirements and assess reporting needs based on client expectations
 - It's important to start here so the scope of the project is clearly defined, documented, and understood fully by testers
- Validate data sources
 - Perform a data count check and verify that the table and column data type meets specifications of the data model
 - If not done correctly, the data in data warehouse, semantic layer, reports could be inaccurate or misleading
- Design test cases
 - Design ETL mapping scenarios, create SQL scripts, and define transformational rules to be executed in parallel with ETL jobs
 - It is important to validate the mapping document as well, to ensure it contains all of the information
- Extract data from source systems
 - Execute ETL tests per business requirement; identify types of bugs or defects encountered during testing and make a report
 - It is important to detect and reproduce any defects, report, fix the bug, resolve, and close bug report before continuing to next step

Testing Process

- Apply transformation logic (ETL flow testing)
 - Ensure data is transformed to match schema of target data warehouse
 - Check data threshold, alignment, and validate data flow
 - This ensures the data type matches the mapping document for each column and table
- Load data into target data warehouse
 - Perform a record count check before and after data is moved from staging to the data warehouse
 - Confirm that invalid data is rejected and that the default values are accepted
- BI report/dashboard testing
 - Verify layout, options, filters and export functionality of BI reports/dashboards
- Performance testing
- Security testing
- Test Closure
- File test closure